

Assignment – Studio3T

ANKITA DAS CSF 017

Data:

```
{ roll: 1, name: "Anu", age: 22, department: "IT", marks: 85 },  
{ roll: 2, name: "Riya", age: 21, department: "CSE", marks: 78 },  
{ roll: 3, name: "Aman", age: 23, department: "IT", marks: 92 },  
{ roll: 4, name: "Neha", age: 20, department: "ECE", marks: 74 },  
{ roll: 5, name: "Raj", age: 24, department: "CSE", marks: 88 },  
{ roll: 6, name: "Simran", age: 22, department: "IT", marks: 81 },  
{ roll: 7, name: "Karan", age: 21, department: "ECE", marks: 69 }, { roll:  
8, name: "Priya", age: 23, department: "CSE", marks: 95 }.
```

Answer : db.students.insertMany([

```
{ roll: 1, name: "Anu", age: 22, department: "IT", marks: 85 },  
{ roll: 2, name: "Riya", age: 21, department: "CSE", marks: 78 },  
{ roll: 3, name: "Aman", age: 23, department: "IT", marks: 92 },  
{ roll: 4, name: "Neha", age: 20, department: "ECE", marks: 74 },  
{ roll: 5, name: "Raj", age: 24, department: "CSE", marks: 88 },  
{ roll: 6, name: "Simran", age: 22, department: "IT", marks: 81 },  
{ roll: 7, name: "Karan", age: 21, department: "ECE", marks: 69 },  
{ roll: 8, name: "Priya", age: 23, department: "CSE", marks: 95 }  
])
```

Q1. Display all documents

```

1 use collegeDBCSECSF_017
2
3 db.students.insertMany([
4 { roll: 1, name: "Anu", age: 22, department: "IT", marks: 85 },
5 { roll: 2, name: "Riya", age: 21, department: "CSE", marks: 78 },
6 { roll: 3, name: "Aman", age: 23, department: "IT", marks: 92 },
7 { roll: 4, name: "Neha", age: 20, department: "ECE", marks: 74 },
8 { roll: 5, name: "Raj", age: 24, department: "CSE", marks: 88 },
9 { roll: 6, name: "Simran", age: 22, department: "IT", marks: 81 },
10 { roll: 7, name: "Karan", age: 21, department: "ECE", marks: 69 },
11 { roll: 8, name: "Priya", age: 23, department: "CSE", marks: 95 }
12 ])
13 db.students.find()
14

```

Raw shell output | Shell Output (Documents) | Find Query (line 13)

50 Documents 1 to 48

students > roll

_id	roll	name	age	department
69cfd796fd8f074...	1	Anu	22	IT
69cfd796fd8f074...	2	Riya	21	CSE
69cfd796fd8f074...	3	Aman	23	IT
69cfd796fd8f074...	4	Neha	20	ECE
69cfd796fd8f074...	5	Raj	24	CSE
69cfd796fd8f074...	6	Simran	22	IT
69cfd796fd8f074...	7	Karan	21	ECE
69cfd796fd8f074...	8	Priya	23	CSE
69cfd7edfd8f074...	1	Anu	22	IT
69cfd7edfd8f074...	2	Riya	21	CSE
69cfd7edfd8f074...	3	Aman	23	IT
69cfd7edfd8f074...	4	Neha	20	ECE
69cfd7edfd8f074...	5	Raj	24	CSE
69cfd7edfd8f074...	6	Simran	22	IT
69cfd7edfd8f074...	7	Karan	21	ECE
69cfd7edfd8f074...	8	Priya	23	CSE
69cfd846827901e...	1	Anu	22	IT
69cfd846827901e...	2	Riya	21	CSE
69cfd846827901e...	3	Aman	23	IT
69cfd846827901e...	4	Neha	20	ECE
69cfd846827901e...	5	Raj	24	CSE

1 document selected | Count Documents | 00:00:00.004

Q2. Find students from IT department

```

1 use collegeDBCSECSF_017
2
3 db.students.insertMany([
4 { roll: 1, name: "Anu", age: 22, department: "IT", marks: 85 },
5 { roll: 2, name: "Riya", age: 21, department: "CSE", marks: 78 },
6 { roll: 3, name: "Aman", age: 23, department: "IT", marks: 92 },
7 { roll: 4, name: "Neha", age: 20, department: "ECE", marks: 74 },
8 { roll: 5, name: "Raj", age: 24, department: "CSE", marks: 88 },
9 { roll: 6, name: "Simran", age: 22, department: "IT", marks: 81 },
10 { roll: 7, name: "Karan", age: 21, department: "ECE", marks: 69 },
11 { roll: 8, name: "Priya", age: 23, department: "CSE", marks: 95 }
12 ])
13 db.students.find()
14 db.students.find({ department: "IT" })

```

Raw shell output | Shell Output (Documents) | Find Query (line 13) | Find Query (line 14)

50 Documents 1 to 21

students > roll

_id	roll	name	age	department
69cfd796fd8f074...	1	Anu	22	IT
69cfd796fd8f074...	3	Aman	23	IT
69cfd796fd8f074...	6	Simran	22	IT
69cfd7edfd8f074...	1	Anu	22	IT
69cfd7edfd8f074...	3	Aman	23	IT
69cfd7edfd8f074...	6	Simran	22	IT
69cfd846827901e...	1	Anu	22	IT
69cfd846827901e...	3	Aman	23	IT
69cfd846827901e...	6	Simran	22	IT
69cfd846827901e...	1	Anu	22	IT
69cfd846827901e...	3	Aman	23	IT
69cfd846827901e...	6	Simran	22	IT
69cfd846827901e...	1	Anu	22	IT
69cfd846827901e...	3	Aman	23	IT
69cfd846827901e...	6	Simran	22	IT
69cfd846827901e...	1	Anu	22	IT
69cfd846827901e...	3	Aman	23	IT
69cfd846827901e...	6	Simran	22	IT
69cfd846827901e...	1	Anu	22	IT
69cfd846827901e...	3	Aman	23	IT
69cfd846827901e...	6	Simran	22	IT

Q3. Find students with marks > 80

1 use collegeDBCSECSF_017	Raw shell output	Shell Output (D...	Find Query (lin...	Find Query (lin...	Find Query (lin...
2					
3 db.students.insertMany([					
4 { roll: 1, name: "Anu", age: 22, department: "IT", marks: 85 },					
5 { roll: 2, name: "Riya", age: 21, department: "CSE", marks: 78					
6 { roll: 3, name: "Aman", age: 23, department: "IT", marks: 92 }					
7 { roll: 4, name: "Neha", age: 20, department: "ECE", marks: 74					
8 { roll: 5, name: "Raj", age: 24, department: "CSE", marks: 88 }					
9 { roll: 6, name: "Simran", age: 22, department: "IT", marks: 81					
10 { roll: 7, name: "Karan", age: 21, department: "ECE", marks: 69					
11 { roll: 8, name: "Priya", age: 23, department: "CSE", marks: 95					
12])					
13 db.students.find()					
14 db.students.find({ department: "IT" })					
15 db.students.find({ marks: { \$gt: 80 } })					
16					

students > roll	roll	name	age	department
69cfd796fd8f074...	1	Anu	22	IT
69cfd796fd8f074...	3	Aman	23	IT
69cfd796fd8f074...	5	Raj	24	CSE
69cfd796fd8f074...	6	Simran	22	IT
69cfd796fd8f074...	8	Priya	23	CSE
69cfd7edfd8f074...	1	Anu	22	IT
69cfd7edfd8f074...	3	Aman	23	IT
69cfd7edfd8f074...	5	Raj	24	CSE
69cfd7edfd8f074...	6	Simran	22	IT
69cfd7edfd8f074...	8	Priya	23	CSE
69cfd846827901e...	1	Anu	22	IT
69cfd846827901e...	3	Aman	23	IT
69cfd846827901e...	5	Raj	24	CSE
69cfd846827901e...	6	Simran	22	IT
69cfd846827901e...	8	Priya	23	CSE
69cff45e827901e...	1	Anu	22	IT
69cff45e827901e...	3	Aman	23	IT
69cff45e827901e...	5	Raj	24	CSE
69cff45e827901e...	6	Simran	22	IT
69cff45e827901e...	8	Priya	23	CSE
69cff45e827901e...	1	Anu	22	IT

Q4. Insert one new student

1 use collegeDBCSECSF_017	Find Query (li...	Find Query (li...	Find Query (li...	Shell Output (D...
2				
3 db.students.insertMany([				
4 { roll: 1, name: "Anu", age: 22, department: "IT", marks: 85 },				
5 { roll: 2, name: "Riya", age: 21, department: "CSE", marks: 78				
6 { roll: 3, name: "Aman", age: 23, department: "IT", marks: 92 }				
7 { roll: 4, name: "Neha", age: 20, department: "ECE", marks: 74				
8 { roll: 5, name: "Raj", age: 24, department: "CSE", marks: 88 }				
9 { roll: 6, name: "Simran", age: 22, department: "IT", marks: 81				
10 { roll: 7, name: "Karan", age: 21, department: "ECE", marks: 69				
11 { roll: 8, name: "Priya", age: 23, department: "CSE", marks: 95				
12])				
13 db.students.find()				
14 db.students.find({ department: "IT" })				
15 db.students.find({ marks: { \$gt: 80 } })				
16 db.students.insertOne(				
17 { roll: 9, name: "Rohit", age: 22, department: "CSE", marks: 82				
18 })				

1 {	"acknowledged" : true,
2 "insertedId" : ObjectId("69cff72c827901e9ec441577")	
3 }	
4 }	
5 }	

Q5. Update marks of Anu to 90

1 use collegeDBCSECSF_017	Find Query (li...	Find Query (li...	Find Query (li...	Shell Output (D...
2				
3 db.students.insertMany([				
4 { roll: 1, name: "Anu", age: 22, department: "IT", marks: 85 },				
5 { roll: 2, name: "Riya", age: 21, department: "CSE", marks: 78				
6 { roll: 3, name: "Aman", age: 23, department: "IT", marks: 92 }				
7 { roll: 4, name: "Neha", age: 20, department: "ECE", marks: 74				
8 { roll: 5, name: "Raj", age: 24, department: "CSE", marks: 88 }				
9 { roll: 6, name: "Simran", age: 22, department: "IT", marks: 81				
10 { roll: 7, name: "Karan", age: 21, department: "ECE", marks: 69				
11 { roll: 8, name: "Priya", age: 23, department: "CSE", marks: 95				
12])				
13 db.students.find()				
14 db.students.find({ department: "IT" })				
15 db.students.find({ marks: { \$gt: 80 } })				
16 db.students.insertOne(				
17 { roll: 9, name: "Rohit", age: 22, department: "CSE", marks: 82				
18 })				
19 db.students.updateOne(				
20 { name: "Anu" },				
21 { \$set: { marks: 90 } }				
22 })				

1 {	"acknowledged" : true,
2 "insertedId" : ObjectId("69cff771827901e9ec441580")	
3 }	
4 }	
5 {	"acknowledged" : true,
6 "insertedId" : null,	
7 "matchedCount" : 1.0,	
8 "modifiedCount" : 1.0,	
9 "upsertedCount" : 0.0	
10 }	
11 }	
12 }	

Q6. Add a new field "status: Pass" for all students

```
1 use collegeDBCSECSF_017
2
3 db.students.insertMany([
4   { roll: 1, name: "Anu", age: 22, department: "IT", marks: 85 },
5   { roll: 2, name: "Riya", age: 21, department: "CSE", marks: 78 },
6   { roll: 3, name: "Aman", age: 23, department: "IT", marks: 92 },
7   { roll: 4, name: "Neha", age: 20, department: "ECE", marks: 74 },
8   { roll: 5, name: "Raj", age: 24, department: "CSE", marks: 88 },
9   { roll: 6, name: "Simran", age: 22, department: "IT", marks: 81 },
10  { roll: 7, name: "Karan", age: 21, department: "ECE", marks: 69 },
11  { roll: 8, name: "Priya", age: 23, department: "CSE", marks: 95 }
12 ])
13
14 db.students.find()
15 db.students.find({ department: "IT" })
16 db.students.find({ marks: { $gt: 80 } })
17 db.students.insertOne(
18   { roll: 9, name: "Rohit", age: 22, department: "CSE", marks: 82 }
19 )
20 db.students.updateOne(
21   { name: "Anu" },
22   { $set: { marks: 90 } }
23 )
24 db.students.updateMany(
25   {},
26   { $set: { status: "Pass" } }
27 )
```

```
Find Query (li... ✖ Find Query (li... ✖ Find Query (li... ✖ Shell
|< < > >| 50 Documents 1 to 3
1 {
2   "acknowledged" : true,
3   "insertedId" : ObjectId("69cff7e3827901e9ec44159b")
4 }
5 {
6   "acknowledged" : true,
7   "insertedId" : null,
8   "matchedCount" : 1.0,
9   "modifiedCount" : 0.0,
10  "upsertedCount" : 0.0
11 }
12 {
13   "acknowledged" : true,
14   "insertedId" : null,
15   "matchedCount" : 117.0,
16   "modifiedCount" : 9.0,
17   "upsertedCount" : 0.0
18 }
19
```

Q7. Delete all students with marks < 70

```
1 use collegeDBCSECSF_017
2
3 db.students.insertMany([
4   { roll: 1, name: "Anu", age: 22, department: "IT", marks: 85 },
5   { roll: 2, name: "Riya", age: 21, department: "CSE", marks: 78 },
6   { roll: 3, name: "Aman", age: 23, department: "IT", marks: 92 },
7   { roll: 4, name: "Neha", age: 20, department: "ECE", marks: 74 },
8   { roll: 5, name: "Raj", age: 24, department: "CSE", marks: 88 },
9   { roll: 6, name: "Simran", age: 22, department: "IT", marks: 81 },
10  { roll: 7, name: "Karan", age: 21, department: "ECE", marks: 69 },
11  { roll: 8, name: "Priya", age: 23, department: "CSE", marks: 95 }
12 ])
13
14 db.students.find()
15 db.students.find({ department: "IT" })
16 db.students.find({ marks: { $gt: 80 } })
17 db.students.insertOne(
18   { roll: 9, name: "Rohit", age: 22, department: "CSE", marks: 82 }
19 )
20 db.students.updateOne(
21   { name: "Anu" },
22   { $set: { marks: 90 } }
23 )
24 db.students.updateMany(
25   {},
26   { $set: { status: "Pass" } }
27 )
28 db.students.deleteMany(
29   { marks: { $lt: 70 } }
30 )
```

```
Find Query (li... ✖ Find Query (li... ✖ Find Query (li... ✖ Shell Output (D...
|< < > >| 50 Documents 1 to 4
1 {
2   "acknowledged" : true,
3   "insertedId" : ObjectId("69cff82c827901e9ec44159b")
4 }
5 {
6   "acknowledged" : true,
7   "insertedId" : null,
8   "matchedCount" : 1.0,
9   "modifiedCount" : 0.0,
10  "upsertedCount" : 0.0
11 }
12 {
13   "acknowledged" : true,
14   "insertedId" : null,
15   "matchedCount" : 126.0,
16   "modifiedCount" : 9.0,
17   "upsertedCount" : 0.0
18 }
19 {
20   "acknowledged" : true,
21   "deletedCount" : 15.0
22 }
23
```

Q8. Count students in CSE department

```

use collegeDBCSECSF_017

db.students.insertMany([
{ roll: 1, name: "Anu", age: 22, department: "IT", marks: 85 },
{ roll: 2, name: "Riya", age: 21, department: "CSE", marks: 78 },
{ roll: 3, name: "Aman", age: 23, department: "IT", marks: 92 },
{ roll: 4, name: "Neha", age: 20, department: "ECE", marks: 74 },
{ roll: 5, name: "Raj", age: 24, department: "CSE", marks: 88 },
{ roll: 6, name: "Simran", age: 22, department: "IT", marks: 81 },
{ roll: 7, name: "Karan", age: 21, department: "ECE", marks: 69 },
{ roll: 8, name: "Priya", age: 23, department: "CSE", marks: 95 }
])
db.students.find()
db.students.find({ department: "IT" })
db.students.find({ marks: { $gt: 80 } })
db.students.insertOne(
{ roll: 9, name: "Rohit", age: 22, department: "CSE", marks: 82 }
)
db.students.updateOne(
{ name: "Anu" },
{ $set: { marks: 90 } }
)
db.students.updateMany(
{ },
{ $set: { status: "Pass" } }
)
db.students.deleteMany(
{ marks: { $lt: 70 } }
)
db.students.countDocuments({ department: "CSE" })

```

```

422   "status" : "Pass"
423 }
424 {
425   "_id" : ObjectId("69cff718827901e9ec44156a"),
426   "roll" : NumberInt(5),
427   "name" : "Raj",
428   "age" : NumberInt(24),
429   "department" : "CSE",
430   "marks" : NumberInt(88),
431   "status" : "Pass"
432 }
433 {
434   "_id" : ObjectId("69cff718827901e9ec44156b"),
435   "roll" : NumberInt(6),
436   "name" : "Simran",
437   "age" : NumberInt(22),
438   "department" : "IT",
439   "marks" : NumberInt(81),
440   "status" : "Pass"
441 }
442 {
443   "_id" : ObjectId("69cff718827901e9ec44156d"),
444   "roll" : NumberInt(8),
445   "name" : "Priya",
446   "age" : NumberInt(23),
447   "department" : "CSE",
448   "marks" : NumberInt(95),
449   "status" : "Pass"
450 }
451 }

```

Q9. Find total students in each department

```

localhost:27017 > collegeDBCSECSF_017

```

```

{ roll: 3, name: "Aman", age: 23, department: "IT", marks: 92 }
{ roll: 4, name: "Neha", age: 20, department: "ECE", marks: 74 }
{ roll: 5, name: "Raj", age: 24, department: "CSE", marks: 88 }
{ roll: 6, name: "Simran", age: 22, department: "IT", marks: 81 }
{ roll: 7, name: "Karan", age: 21, department: "ECE", marks: 69 }
{ roll: 8, name: "Priya", age: 23, department: "CSE", marks: 95 }
])
db.students.find()
db.students.find({ department: "IT" })
db.students.find({ marks: { $gt: 80 } })
db.students.insertOne(
{ roll: 9, name: "Rohit", age: 22, department: "CSE", marks: 82 }
)
db.students.updateOne(
{ name: "Anu" },
{ $set: { marks: 90 } }
)
db.students.updateMany(
{ },
{ $set: { status: "Pass" } }
)
db.students.deleteMany(
{ marks: { $lt: 70 } }
)
db.students.countDocuments({ department: "CSE" })
db.students.aggregate([
{
  $group: {
    _id: "$department",
    totalStudents: { $sum: 1 }
  }
}
])

```

```

1 {
2   "_id" : "CSE",
3   "totalStudents" : 59.0
4 }
5 {
6   "_id" : "IT",
7   "totalStudents" : 51.0
8 }
9 {
10  "_id" : "ECE",
11  "totalStudents" : 17.0
12 }
13 }

```

Q10. Find highest marks

```

7 { roll: 4, name: "Neha", age: 20, department: "ECE", marks: 78 }
8 { roll: 5, name: "Raj", age: 24, department: "CSE", marks: 88 }
9 { roll: 6, name: "Simran", age: 22, department: "IT", marks: 92 }
10 { roll: 7, name: "Karan", age: 21, department: "ECE", marks: 85 }
11 { roll: 8, name: "Priya", age: 23, department: "CSE", marks: 95 }
12 }
13 db.students.find()
14 db.students.find({ department: "IT" })
15 db.students.find({ marks: { $gt: 80 } })
16 db.students.insertOne(
17 { roll: 9, name: "Rohit", age: 22, department: "CSE", marks: 85 }
18 )
19 db.students.updateOne(
20 { name: "Anu" },
21 { $set: { marks: 90 } }
22 )
23 db.students.updateMany(
24 {},
25 { $set: { status: "Pass" } }
26 )
27 db.students.deleteMany(
28 { marks: { $lt: 70 } }
29 )
30 db.students.countDocuments({ department: "CSE" })
31 db.students.aggregate([
32 {
33   $group: {
34     _id: "$department",
35     totalStudents: { $sum: 1 }
36   }
37 }
38 ])
39 db.students.find().sort({ marks: -1 }).limit(1)

```

```

1 {
2   "_id" : ObjectId("69cfd796fd8f07461c441535"),
3   "roll" : NumberInt(8),
4   "name" : "Priya",
5   "age" : NumberInt(23),
6   "department" : "CSE",
7   "marks" : NumberInt(95),
8   "status" : "Pass"
9 }
10

```

Q12. Find students whose name starts with 'A' and marks > 80

```

13 db.students.find()
14 db.students.find({ department: "IT" })
15 db.students.find({ marks: { $gt: 80 } })
16 db.students.insertOne(
17 { roll: 9, name: "Rohit", age: 22, department: "CSE", marks: 85 }
18 )
19 db.students.updateOne(
20 { name: "Anu" },
21 { $set: { marks: 90 } }
22 )
23 db.students.updateMany(
24 {},
25 { $set: { status: "Pass" } }
26 )
27 db.students.deleteMany(
28 { marks: { $lt: 70 } }
29 )
30 db.students.countDocuments({ department: "CSE" })
31 db.students.aggregate([
32 {
33   $group: {
34     _id: "$department",
35     totalStudents: { $sum: 1 }
36   }
37 }
38 ])
39 db.students.find().sort({ marks: -1 }).limit(1)
40 db.students.find().sort({ marks: -1 }).skip(1).limit(1)
41 db.students.find(
42 { name: { $regex: "A" },
43   marks: { $gt: 80 } }
44 )

```

```

182 {
183   "_id" : ObjectId("69cfd796fd8f07461c441535"),
184   "roll" : NumberInt(1),
185   "name" : "Anu",
186   "age" : NumberInt(22),
187   "department" : "IT",
188   "marks" : NumberInt(85),
189   "status" : "Pass"
190 }
191 {
192   "_id" : ObjectId("69cfd796fd8f07461c441535"),
193   "roll" : NumberInt(3),
194   "name" : "Aman",
195   "age" : NumberInt(23),
196   "department" : "IT",
197   "marks" : NumberInt(92),
198   "status" : "Pass"
199 }
200 {
201   "_id" : ObjectId("69cfd796fd8f07461c441535"),
202   "roll" : NumberInt(1),
203   "name" : "Anu",
204   "age" : NumberInt(22),
205   "department" : "IT",
206   "marks" : NumberInt(85),
207   "status" : "Pass"
208 }
209 {
210   "_id" : ObjectId("69cfd796fd8f07461c441535"),
211   "roll" : NumberInt(3),
212   "name" : "Aman",
213   "age" : NumberInt(23),
214   "department" : "IT",
215   "marks" : NumberInt(92),
216   "status" : "Pass"
217 }

```

Q13. Display department-wise average marks, only for departments where avg > 80

The screenshot shows the MongoDB Shell interface with a script being executed. The script performs the following operations:

- Update many documents in the `students` collection, setting `status` to `"Pass"`.
- Delete many documents from the `students` collection where `marks` is less than 70.
- Count documents in the `students` collection where `department` is `"CSE"`.
- Aggregate documents to calculate the average marks for each department.
- Find documents sorted by marks in descending order, limited to 1 document.
- Find documents sorted by marks in descending order, skipping 1 document and limiting to 1 document.
- Find documents where the name matches a regular expression `"^A"` and marks are greater than 80.
- Aggregate documents to calculate the average marks for each department.
- Match documents where the average marks are greater than 80.

The right pane shows the JSON view of the results, displaying two documents with their `_id`, `avgMarks`, and `status`.

Q14. Find students with marks between 75 and 90, sorted by marks ascending Q15. Add grade field based on marks

The screenshot shows the MongoDB Shell interface with a script being executed. The script performs the following operations:

- Delete many documents from the `students` collection where `marks` is less than 70.
- Count documents in the `students` collection where `department` is `"CSE"`.
- Aggregate documents to calculate the average marks for each department.
- Find documents sorted by marks in descending order, limited to 1 document.
- Find documents sorted by marks in descending order, skipping 1 document and limiting to 1 document.
- Find documents where the name matches a regular expression `"^A"` and marks are greater than 80.
- Aggregate documents to calculate the average marks for each department.
- Match documents where the average marks are greater than 80.
- Find documents where marks are greater than or equal to 75 and less than or equal to 90, sorted by marks in ascending order.

The right pane shows the JSON view of the results, displaying three documents with their `_id`, `roll`, `name`, `age`, `department`, `marks`, and `status`.

Q15. Add grade field based on marks

localhost27017 > collegeDBCSECSF_017

Find Query (li... Aggregate Quer... Find Query (li... Shell Output (D...
50 Documents 1 to 1 JSON View

```
45 db.students.aggregate([
46 {
47   $group: {
48     _id: "$department",
49     avgMarks: { $avg: "$marks" }
50   }
51 },
52 $match: { avgMarks: { $gt: 80 } }
53 })
54 db.students.find(
55 marks: { $gte: 75, $lte: 90 }
56 ).sort({ marks: 1 })
57 db.students.updateMany(
58 {
59   $set: {
60     grade: {
61       $switch: {
62         branches: [
63           { case: { $gte: ["$marks", 90] }, then: "A" },
64           { case: { $gte: ["$marks", 80] }, then: "B" },
65           { case: { $gte: ["$marks", 70] }, then: "C" }
66         ],
67         default: "D"
68       }
69     }
70   }
71 ]
72 }
```

```
1 {
2   "acknowledged" : true,
3   "insertedId" : null,
4   "matchedCount" : 215.0,
5   "modifiedCount" : 215.0,
6   "upsertedCount" : 0.0
7 }
8
```

1 document selected 00:00:00.626